



Collision avoidance for an unmanned surface vehicle using deep reinforcement learning

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ABSTRACT

In this paper, a deep reinforcement learning (DRL)-based collision avoidance method is proposed for an unmanned surface vehicle (USV). This approach is applicable to the decision-making stage of collision avoidance, which determines whether the avoidance is necessary, and if so, determines the direction of the avoidance maneuver. To utilize the visual recognition capability of deep neural networks as a tool for analyzing the complex and ambiguous situations that are typically encountered, a grid map representation of the ship encounter situation was suggested. For the composition of the DRL network, we proposed a neural network architecture and semi-Markov decision process model that was specially designed for the USV collision avoidance problem. The proposed DRL network was trained through repeated simulations of collision avoidance. After the training process, the DRL network was implemented in collision avoidance experiments and simulations to evaluate its situation recognition and collision avoidance capability.

1. Introduction

1.1. Research background

In recent years, as the social demands for unmanned systems have increased and the performance and reliability of related technologies have improved, human-operated systems have been actively replaced by unmanned systems. This trend is also apparent in the maritime domain. In the case of the unmanned surface vehicle (USV), the unmanned system can substitute for human operations that are time consuming or dangerous such as environmental monitoring or mine searching.

Several different types of USVs have been recently developed or are under development. For example, Defense Advanced Research Projects Agency (DARPA) developed a USV for long-term tracking of enemy submarines (Sea hunter), while an Israeli military company has developed a USV for reconnaissance of the Persian Gulf coast (Protector). In the private sector, a USV has been developed for long-term oceanic data collection (C-enduro), as well as marine transportation (Revolt). Although these USVs perform different tasks and were developed for different purposes, they must have the ability to avoid collisions to nearby obstacles to successfully perform their task. Collisions can cause structural damage in surface vehicles. Moreover, of equal importance is the risk of property damage, environmental pollution caused by oil spillage, and the loss of life. As such, USVs must possess collision

avoidance capabilities that are equivalent to that of a human operator or beyond.

According to Campbell et al. (2012), investigations have revealed that most ship collision accidents are caused by human errors such as guard negligence or violation of collision prevention rules. In the case of USV, we can exclude human error from the collision avoidance process. Therefore, it is expected that the collision accident probability in a marine environment can be significantly reduced by the development and deployment of effective and reliable USV collision avoidance algorithms.

1.2. Previous research

In this research, we adopted the deep reinforcement learning (DRL) approach to deal with USV collision avoidance problem. The following is a representative example of precedent research for both DRL and ship collision avoidance. Research in the field of ship collision avoidance can be classified into obstacle detection, inference of the time of collision avoidance, and avoidance path planning.

1.2.1. Obstacle detection

To detect nearby obstacles, USV depends on perception sensors such as radar, LiDAR (light detection and ranging), vision sensor, sonar

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and etc. Almeida et al. (2009) conducted a study on the detection of obstacles using a mounted radar and assessed the risk of collision using the closest point of approach (CPA) information. Woo and Kim (2015) have conducted research on vision-based obstacle detection of USV using monocular cameras. In order to determine the region of interest in an image, a horizontal line was extracted using the RANSAC algorithm and feature extraction methods were used to detect obstacles. Lebbad and Nataraj (2015) used LiDAR sensors to estimate the obstacle region of interest in a vision image. According to this work, they determined that there is distortion of the LiDAR image due to the oscillatory motion induced by environmental disturbances such as waves.

1.2.2. Inference of the time of collision avoidance

Research on the inference of the time of collision avoidance has been actively pursued since the 1970s. Previous research on this topic can be categorized into approaches based on the ship domain and the closest point of approach (CPA) methods.

The ship domain is a concept proposed by Fujii and Tanaka (1971) that establishes a virtual safety zone around a vessel and performs collision avoidance maneuvers when any obstacle violates the virtual zone. Subsequent to this work, several researchers such as Goodwin (1975), Coldwell (1983) and Davis et al. (1980) proposed an advanced ship domain that reflects the COLREGs in the formation of the ship domain. In these works, a lateral asymmetry ship domain was proposed to allocate a greater area to the starboard side of the vessel.

The collision avoidance time inference method using CPA was initially proposed by Iwasaki and Hara. This approach estimates collision risk based on distance to closest point of approach (DCPA) and time to closest point of approach (TCPA) information. Hasegawa and Kouzuki (1987) attempted to reflect the experience of an expert ship operator by designing membership function and rules of a fuzzy estimator based on interviews with an expert.

1.2.3. Avoidance path planning

Ship collision avoidance path planning algorithms can be categorized into two groups; global path planning and local path planning. Global path planning methods generate collision avoidance paths based on an obstacle map information. However, a local path planning method generates local avoidance path based on real-time sensor information.

As a global path planning technique, there are path generation approaches based on the A* algorithm for the grid map proposed by Larson et al. (2006), and the visibility graph (VG) proposed by Casalino et al. (2009).

A representative example of the local path planning technique for the USV is the collision avoidance technique using the velocity obstacle (VO) proposed by Kuwata et al. (2014). This approach has significantly influenced subsequent research by Myre (2016) and Stenersen (2015). In the VO technique, the obstacle represents a set of velocities that cause a collision when considering the kinematics of moving obstacles and the vehicle. By selecting velocity and course angles outside of the VO, the vehicle can avoid collision with nearby obstacles.

1.2.4. Deep reinforcement learning

In recent years, the deep reinforcement learning method have been actively adopted to deal with various control problem in robotic domain. Polvara et al. applied the DRL method to end-to-end autonomous landing of an UAV (unmanned aerial vehicle) toward stationary marker (Polvara et al., 2018), as well as to a deck of an USV platform (Polvara et al., 2019).

Similarly, a number of researcher applied the DRL method to deal with guidance and control of mobile robot (Zhu et al., 2017; Tai et al., 2017; Kahn et al., 2018), especially in the indoor environment. Due to the flexibility of the input layer configuration, input data of the DRL network are varied from visual image (Zhu et al., 2017; Kahn et al., 2018) to laser sensor signal (Tai et al., 2017).

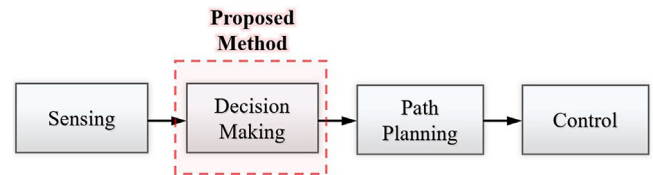


Fig. 1. The stages involved in autonomous surface vehicle collision avoidance.

Because of its control capability of the complex and coupled system, the DRL method is often used to control manipulator robots as well (Kahn et al., 2018). In this work, Gu et al. (2017) validated the effectiveness of asynchronous off-policy DRL algorithm can successfully perform complex manipulation task such as *target reaching* and *door opening*.

The remainder of this report is organized in sections. In Section 2, the background concepts used in this work with respect to ship collision avoidance are presented. In Section 3, the background and problem formulation of the deep reinforcement learning based collision avoidance algorithm is given. In Section 4, validation of the proposed deep reinforcement learning based collision avoidance method is examined. A feasibility analysis, in addition to simulation- and experiment-based collision avoidance test results are also presented in this section. Finally, in Section 5, the conclusions of the research and further discussion are reviewed.

2. Ship collision avoidance

2.1. Stages of ship collision avoidance

The process of collision avoidance in USVs generally involves four steps shown as Fig. 1. In the sensing phase, obstacle detection is performed using perception sensors such as radar, AIS, LiDAR and vision sensors. Using the perceived information, motion information of the obstacle is obtained. At this point, a collision avoidance decision is made during the decision making phase. In this stage, the USV decide whether the collision avoidance action is necessary or not, and if necessary, determines the appropriate avoidance action. If collision avoidance action is necessary, the USV enters the path planning phase. In this step, a path planner determines the desired guidance command to attempt avoidance action. In the last step, a controller calculates the desired control input for each actuator to perform the collision avoidance action based on the guidance command generated during the path planning stage.

Our work is focused on the decision-making stage. However, in the process of performing collision avoidance simulations and experiments for USVs, the other three stages must be considered. As such, we adopted the velocity obstacle path planning method to for the path planning stage and for the controller, we designed a proportional–integral–differential (PID) based steering and speed controller to track the guidance command.

In the decision-making stage of collision avoidance, an international regulation for ship collision avoidance should be considered to determine the appropriate avoidance action based on the encountered situation. In the remaining parts of this section, the background concept of the regulation will be presented.

2.2. COLREGs

International regulations for the prevention of collisions at sea (COLREGs) is a set of rules established by the International Maritime Organization (IMO) in 1972 that are mandatory for the operation of marine vessels. COLREGs specifies the give-way vessel and the stand-on vessel of each encountered situation, as well as the desired direction

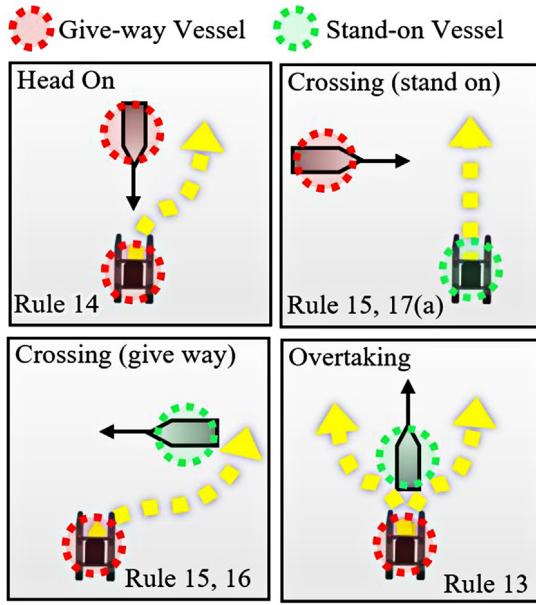


Fig. 2. Desired avoidance direction and give-way vessel for each encountered situation, according to COLREGs.

of avoidance to prevent collisions. Since COLREGs is an internationally accepted set of rules that are closely related to international maritime laws, it is necessary to develop an action plan for unmanned vessels based on COLREGs to ensure safety at sea. The following are representative rules of COLREGs that are often used for ship collision avoidance in a variety of encountered situations (see Fig. 2).

Head on situation

In a head-on encounter situation where the own vessel and the obstacle vessel approach each other, both vessels have a duty to avoid each other by performing a turning maneuver to the starboard side. (COLREGs Article 14)

Crossing (give way)

In a crossing (give way) situation, the obstacle vessel crosses from the starboard side of the own vessel. In this situation, the obstacle vessel is the stand-on vessel, which has no obligation to avoid a collision. The own vessel must perform appropriate avoidance action (give way vessel) to avoid a potential collision. According to COLREGs, the own vessel must not cross the moving direction of the other vessel. In this case, the own vessel must turn to the starboard side to avoid a collision. (COLREGs Article 15)

Crossing (stand on)

In a crossing (stand on) situation, the obstacle vessel crosses from the port side of the own vessel. In this situation, the obstacle vessel is the give-way vessel and must perform appropriate action to avoid a potential collision. However, if the give way vessel does not take appropriate action, the own vessel should perform an appropriate avoidance action to prevent the collision despite being the “stand on” vessel. (COLREGs Article 17 (a))

Overtaking

In an overtaking situation where the own vessel overtakes the obstacle vessel, the former is the give way vessel and obstacle vessel is the stand on vessel. The COLREGs rule for overtaking (Article 13) does not specifically state any desired avoidance direction. Therefore, either the port or the starboard direction of avoiding action is permitted.

2.3. Velocity obstacle path planning

In this study, the VO algorithm was utilized for the planning of the avoidance path. The VO method is commonly used in the field of

robotics to plan safe paths for obstacle collision avoidance. In 1989, Tychonievich proposed the concept of maneuvering board for ship navigation. Thereafter, the technique was developed and improved under different names such as collision cone, velocity map, and velocity obstacle. In recent years, Stenersen, Myre, and Kuwata adopted the VO method in collision avoidance of an unmanned surface vehicle. In the velocity obstacle method, a velocity obstacle, VO , is a set of velocities that cause collisions when considering the relative velocities of moving obstacles among the velocities that can be selected. This can be expressed as mathematical equations where the operation $\oplus$ refers to the Minkowski sum operation ($A \oplus B = \{a + b \mid a \in A, b \in B\}$) and the operation $-$ indicates the reflection operation ($-A = \{-a \mid a \in A\}$)

$$VO_B^A(\vec{v}_B) = \{\vec{v}_A \mid \lambda(\vec{p}_A, \vec{v}_A - \vec{v}_B) \cap (B \oplus -A) \neq \emptyset\} \quad (1)$$

According to Eq. (1), if we assume that the obstacle is defined as disk-shaped, then Eq. (1) can be expressed as Eq. (2), where the $D(x, r)$ represents a disk with radius r and center location vector x .

$$VO_B^A(\vec{v}_B) = \{\vec{v}_A \mid \lambda(\vec{p}_A, \vec{v}_A - \vec{v}_B) \in D(\vec{p}_B - \vec{p}_A, r_{AB})\} \quad (2)$$

Once the velocity obstacle set is calculated using Eq. (2), by selecting a velocity vector outside of the VO and velocity set that violates the COLREGs, a safe guidance command can be calculated. In the work of Kuwata et al. (2014), an optimization technique in a velocity space grid was utilized to determine the best speed and course angle.

3. Deep reinforcement learning-based collision avoidance

3.1. Deep Reinforcement Learning(DRL)

According to Sutton (1992), reinforcement learning (RL) is defined as “the learning of a mapping from situations to actions to maximize a scalar reward or reinforcement signal.” In the reinforcement learning problem, an agent can recognize environmental situations using a state evaluation function called a value function, which is often approximated by a function approximator such as a neural network. In recent years, multilayer neural networks have often been utilized to solve the RL problem, which is known as DRL. Therefore, the domain of the RL problem was extended to a more complex environment including visual image-based decision-making problems because of the deep neural network’s powerful capability in image recognition and visual feature extraction. In 2013, Mnih et al. (2015) proposed a powerful deep RL algorithm known as DQN that significantly increased the stability and speed of the learning process. In the DQN model, the agent is trained to select an optimal action that maximizes cumulative future reward based on its policy π . Similar to other Q-learning based RL algorithm, the optimal action is determined by selecting an action with the optimal action-value function $Q^*(s, a)$, where s and a represent the current state and action, and r_t is a reward value at time t .

$$Q^*(s, a) = \max_{\pi} \mathbb{E} \left[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots \mid s_t = s, a_t = a, \pi \right] \quad (3)$$

During the training process of the DQN, the agent collects experience data sets and they are used to update the action value function so that the result of the agent–environment interaction can be reflected in the action value function $Q(s, a)$. An iterative update process of the action value function is performed using the following Bellman equation, where s' and a' represent the state and action at the next time-step, respectively.

$$Q_{t+1}(s, a) = \max_{a'} \mathbb{E} \left[r + \gamma \max_{a'} Q_t(s', a') \mid s, a \right] \quad (4)$$

Using the convolutional neural network with the parameter θ , the action value function $Q(s, a)$ can be approximated as $Q(s, a; \theta)$ and the action value network can be iteratively updated using the loss function $L(\theta)$, where θ_t is a parameter for the separate deep neural network for target action value.

$$L(\theta) = \mathbb{E}_{(s, a, r, s')} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta_t) - Q(s, a; \theta) \right)^2 \right] \quad (5)$$

After Mnih et al. introduced several technique to increase the stability and performance of the DRL such as *experience replay* and *separate target Q network* used in DQN, several researches have produced works on the enhancement of the performance of DRL. For example, Van Hasselt et al. (2016) proposed a method called Double DQN, which can suppress Q-value overestimation tendency during the training by decoupling the action selection process from the target Q-value estimation process. In the double DQN method, two separate networks were used to determine the best action and to calculate the target Q value respectively. Another import structure is called Dueling DQN, also known as DDQN, suggested by Wang et al. (2015). In this method, the learning process of the action value function (Q-value function) is divided into the value of the current state V and the advantage of the action A as described by the following equation.

$$Q(s, a) = V(s) + A(s, a) \quad (6)$$

By dividing the value function and advantage, learning of V and A is performed in a parallel manner. A benefit of this approach is that the network can separately learn the utility of a state and that of an action so that it can accurately estimate the value function.

To address the ship collision avoidance decision-making problem, we adopted the DQN architecture proposed by Mnih et al. (2015), as well as the double DQN (Van Hasselt et al., 2016) and dueling DQN (Wang et al., 2015) to build the network architecture. A more detailed description of the proposed network architecture will be provided in Section 3.2.

3.2. Ship collision avoidance using DRL

Fig. 3 illustrates the overall block diagram of the proposed collision avoidance system. The first group of blocks (yellow) addresses the recognition of the encountered situation and the selection of the desired behavior of the USV according to the current status. This includes decision-making on whether the vehicle should avoid the obstacle or not, as well as the determination of which direction to avoid. To make appropriate decisions, the DRL network should be able to recognize nearby encounter situations. To identify complex collision features in the encountered situation, we proposed a visual representation method. Block (1) of Fig. 3 generates an encounter situation image and delivers it to the DRL network. In this regard, kinematic information (both position η and velocity v) of the USV is necessary as well as path and obstacle information. When obstacles are present near the USV (determined using TCPA, DCPA comparison in (2) of Fig. 3), the DRL network is activated to perform collision avoidance decision-making. Based on the encountered situation, the DRL network selects one of three behavior candidates from the options of path following, starboard-side avoidance, and port-side avoidance. Once the DRL network designates the desired guidance law as illustrated by the red shaded region of Fig. 3, the corresponding guidance commands (desired course angle χ_d and velocity v_d) can be calculated. Using the guidance command as a tracking reference, the control input u can be determined using steering and speed controllers. Later in this section, a detailed description of the proposed DRL based collision avoidance system will be provided.

3.2.1. Semi-Markov decision process

For the mathematical model of decision-making, we adopted a semi-Markov decision process (SMDP) proposed by Sutton et al. (1999). In SMDP, the concept of action is temporally extended from that of the Markov decision process (MDP), thus the SMDP model can address more sophisticated behavior. The action behavior of the proposed DRL network is defined as high-level behaviors such as path following, starboard avoidance, and port avoidance instead of instant low-level action such as steering control command. By selecting the SMDP as a base model, we can restrict the DRL network's role as a behavior selection, isolate it from the calculation of control command. Because of this approach, the decision making and control processes can be

separated, which enables the systematic analysis of the DRL based simulation or experiment. Besides, mathematical stability analysis of the low-level controller can be conducted by selecting the conventional control algorithms such as PID or LQR controller instead of direct control from deep neural network.

To address the collision avoidance problem, we formulated components of the SMDP (state, action, and reward) in the following manner; firstly, the state of the SMDP should include all the necessary information to judge collision risk and situation. This information includes the target path, relative position, and velocity of obstacles, and own vehicle. In this work, we suggested a state representation of a visual grid map (obstacle image) to represent the geometric information of the encountered situation. As such, we can provide more intuitive information to the DRL network and exploit the deep neural network's image recognition capability to understand and assess encountered situations.

Fig. 4 illustrates an encountered situation and the corresponding grid map representation. The grid map is composed of three layers that contain information on the target path, dynamic (moving) obstacle, and static obstacle respectively. In this work, instead of merging all information into a single channel, we intentionally separated each layer in order to prevent it from mixed or overlapped by other layer information. According to Fig. 4, there exists a body fixed virtual window (336×336 m sized) fixed to the vehicle and any features (obstacle or target path) inside this window is represented in the corresponding grid map layer (84×84 pixel sized) using an intensity value between 0 and 255. For simplicity of representation of the obstacle using a grid map both the static and dynamic obstacles are assumed to have a circular shape as expressed in Fig. 4. In the case of the path layer, a relative path representation inside of the virtual window is expressed.

Given that the static obstacle and target path are stationary, the provision of relative positional information is sufficient to facilitate avoidance decisions. However, for dynamic obstacles, the velocity of the dynamic obstacle should be considered in the avoidance decision process. To express the velocity information, a line that represents the direction of the obstacle's velocity is included as shown in Fig. 4. In the dynamic obstacle layer, the length of the line is defined as being proportional to the speed of the corresponding obstacle. As such, if a target obstacle moves with a high speed, a relatively long line will be represented in the dynamic obstacle layer so that the DRL network can recognize the temporal dependent information using a single image layer.

Although the grid map represents path and nearby obstacle information, some of the own vehicle statuses cannot be expressed using the grid map. This information includes speed, angular rate, and the thruster RPM of the own vehicle. In the problem of USV collision avoidance, such information can be directly accessible from the on-board devices installed on the USV, such as gyro, GPS, and control computer. To feed these information to the network, an additional low-dimensional state vector is utilized. In the work of Mnih et al. a stack of 4 consecutive images was used as an input state to implicitly express the temporal information. However, in our configuration, there is no need for stacking a series of images, since the temporal information is already provided to DRL network by using the low-dimensional state input. By combining these two kinds of states (both the grid map and the low-dimensional state vector), the state of the SMDP represents the current situation.

As a result of the collision avoidance decision-making process, the proposed DRL network selects the desired behavior of the vehicle in an encountered situation and the guidance command is calculated from the corresponding guidance law. We define three behavior candidates; path following, starboard-side avoidance, and port-side avoidance. For the path following behavior, the vector field guidance (VFG) method proposed by Nelson et al. (2007) is utilized to determine the guidance command. Fig. 5(a) illustrates a schematic description of the VFG

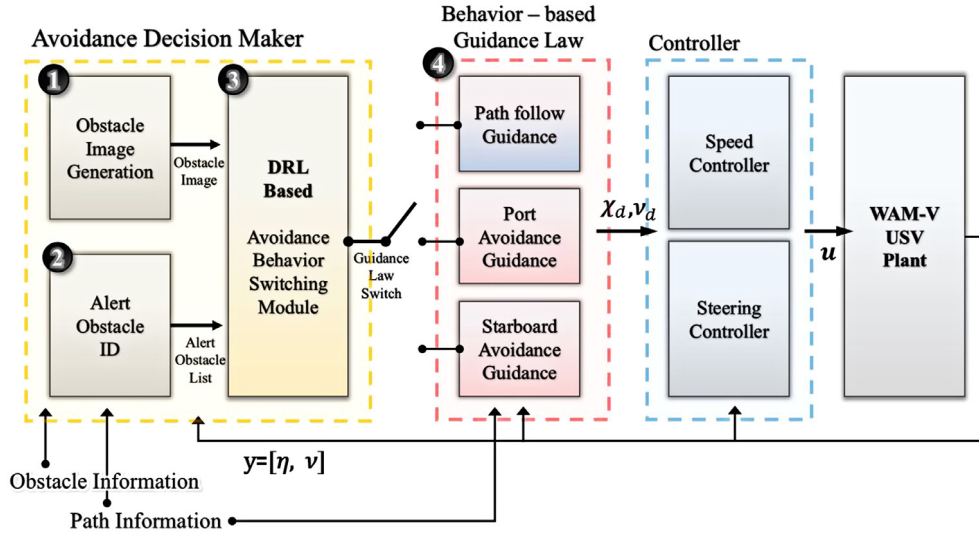


Fig. 3. Schematic diagram of the proposed deep reinforcement learning based collision avoidance system. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

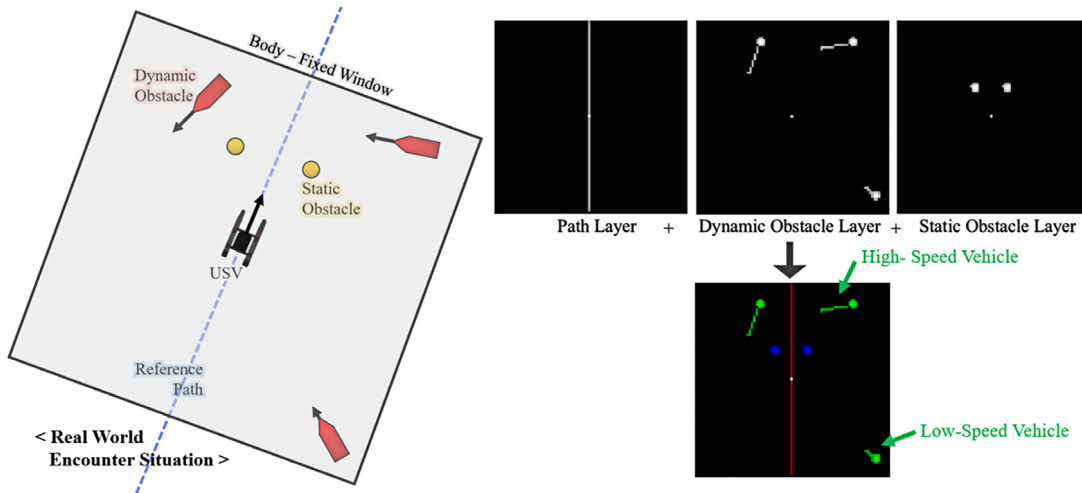


Fig. 4. Schematic description of the representation method for the proposed state(encountered situation) using layers of a grid map.

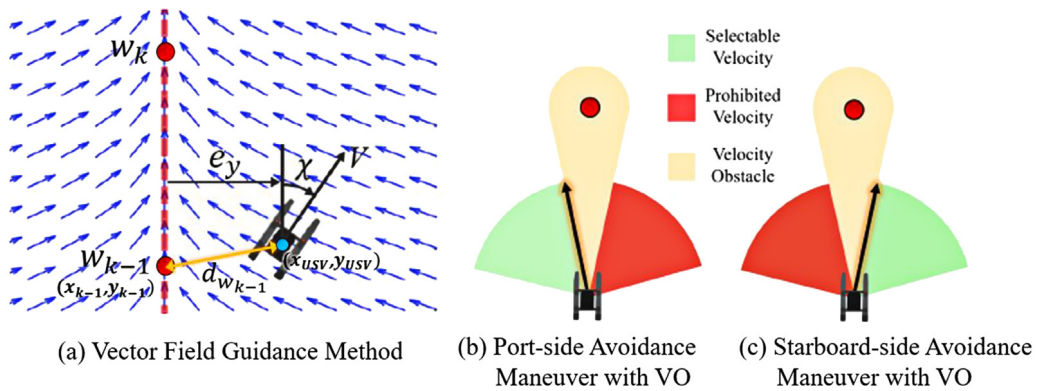


Fig. 5. Guidance methods used for the three behavior candidates of the SMDP.

method, while Eq. (7) represents the desired course angle command calculated using the VFG method. In the equation, χ^∞ and k represent the maximum course angle variation for path following and convergence rate tuning parameters respectively. χ^{path} represents the direction of the target path defined in the inertial reference frame. The cross track

error e_y of the vehicle to the target path can be calculated using the Eq. (9)

$$\chi^d = \chi^\infty \cdot \tan^{-1}(ke_y) + \chi^{path} \quad (7)$$

$$d_{W_{k-1}} = \sqrt{(y_{usv} - y_{k-1})^2 + (x_{usv} - x_{k-1})^2} \quad (8)$$

$$e_y = \sin\left(\chi_{path} - \text{atan}\left(\frac{y_{usv} - y_{k-1}}{x_{usv} - x_{k-1}}\right)\right) \cdot d_{W_{k-1}} \quad (9)$$

For avoidance path planning, we used the VO approach described in Section 2.3. To include the avoidance direction selection process in the decision-making process, we separated the different avoidance direction as independent behaviors. In this regard, we defined separate prohibited velocity regions in the VO for each avoidance direction as illustrated in Fig. 5(b) and (c). The desired speed and course angle can then be selected from the remaining safe velocity space.

After the desired behavior is performed, it should be evaluated using the reward function to train the policy of the DRL network. To address the ship collision avoidance problem, the reward function of the SMDP was defined as follows. We established that there exists three types of rewards for path following, collision avoidance and penalty denoted as r_{path} , r_{avoid} and $r_{penalty}$ respectively. The reward for path following r_{path} is a positive reward that is inverse proportional to the cross track error of the vehicle towards the target path (e_y). r_{path} is included in the reward composition in order to prevent unnecessary or premature avoidance maneuvering of the vehicle. r_{path} is defined by the following equation where $k_{r_{path}}$ is a parameter to tune the sensitivity. For the case when the DRL network conducts path following behavior, the final reward r is defined as $r = r_{path}$.

$$r_{path} = \exp(k_{r_{path}} \cdot \text{abs}(e_y)) \quad (10)$$

Due to the positive reward of path following, the vehicle tends to follow a path when the collision risk is relatively low. In addition, by imposing a positive reward for appropriate collision avoidance maneuvering r_{avoid} , the vehicle can be guided to perform desirable avoidance maneuvers when a collision is imminent. The important aspect of this process is that the efficacy of the avoidance maneuver varies according to the starting time of the collision avoidance. If the avoidance procedure is performed too late, the vehicle may encounter a risky situation or fail to avoid the collision. In contrast, if the procedure is implemented too early, the vehicle may perform unnecessary avoidance maneuvers, although it is still safe to follow the target path. The suitable moment to initiate an avoidance action differs for each situation encountered. Thus, a reward function should be appropriately designed to facilitate the capture of the required moment by the agent for conducting an avoidance maneuver. To achieve this requirement, we have categorized encounter situation into two groups. In the first group, it is assumed that the vehicle should perform avoidance maneuvers as soon as the obstacle is recognized, according to the CPA information (as illustrated in block (3) of Fig. 3). This is the case where the crossing (give way) and overtaking situations should be included because of the give way status of the own vessel. In such cases, it is important to perform instant avoidance motion to provide prompt avoidance intention to the obstacle vehicle. In this case, the reward for the avoidance maneuver r_{avoid} is calculated using the following equation where $k_{r_{avoid}}$ is the sensitivity tuning parameter and $TCPA_{current}$ refers to TCPA at the instance of avoidance. The $TCPA_{thresh}$ is defined as the threshold TCPA value used in block (2) of Fig. 3 to determine the conservative list of the alert obstacle.

$$r_{avoid} = \exp(k_{r_{avoid}} \cdot \text{abs}(TCPA_{current} - TCPA_{thresh})) \quad (11)$$

For the other group, we assumed that the own vehicle should hold on to avoid until achieves the appropriate avoidance time. This is the case for the head on crossing (stand on) and the static obstacle is included. In comparison with the first group, this group of the encounter situation has a relatively large temporal margin until the own vehicle performs avoidance. For the case of the head on situation, both the own vehicle and the obstacle vehicle are give-way vessels. Thus, instead of performing an instant avoidance maneuver, we concluded that it is more important to provide clear and obvious avoidance intention to the obstacle vehicle by performing a large course angle change. When collisions are avoided using the VO method, if the avoidance maneuver is performed too early, it tends to be ambiguous due to

the small change in the course angle. Myre (2016) noted that this characteristic of the VO method degrades the quality of the avoidance process. Thus, in this work, we determined that the avoidance moment during the head on situation should produce 30 degrees of course angle change to provide clear avoidance intention to the obstacle ship. In crossing (stand on) encountered situations, COLREGs suggest that the own vehicle should maintain its course unless it is evident that the stand-on vessel does not perform proper avoidance maneuver. In this case, despite the status of the stand-on vessel, the own vehicle must perform appropriate avoidance maneuvers to avert a catastrophic accident. To reflect this regulation, we determined that the avoidance moment during the crossing (stand on) situation should produce 75 degrees of course angle change. As such, the vehicle in the crossing (stand on) receives a maximum reward value when the implemented course angle change of the avoidance maneuver is 75 degrees. It is the domain of the DRL learning process to identify which temporal moment of the avoidance results in the maximum reward. For the case of the static obstacle, we determined that the avoidance moment during the encountered situation should produce 45 degrees of course angle change. Such thresholds for the course angle change for different situations are determined by trial and error for several collision avoidance simulations. In this group, the reward for avoidance maneuver r_{avoid} is calculated using the following equation where $\hat{\chi}_{avoid}$ represents the change of the course angle due to the avoidance maneuver and $\hat{\chi}_d$ refers to the desired change in the course angle value according to the corresponding situation. For the case when the DRL network conducts avoidance maneuver, the final reward r is defined as $r = r_{avoid}$.

$$r_{avoid} = \exp(k_{r_{avoid}} \cdot \text{abs}(\hat{\chi}_{avoid} - \hat{\chi}_d)) \quad (12)$$

In contrast to the positive reward previously mentioned, there is also a negative reward for penalty term. This penalty reward is active when the vehicle collides with any obstacle or violates the COLREGs rule during the avoidance maneuver. The reward for penalty is defined as the minimum reward value -1 , thus $r_{penalty} = -1$. If any obstacle collision or COLREGs violation event occurs, the reward for the behavior performed is defined as $r = r_{penalty}$ and the learning episode is automatically terminated.

3.2.2. Deep reinforcement learning network

Fig. 6 illustrates the structure of the proposed DRL network. In order to suppress the overestimation phenomenon and stabilize the learning process, we have adopted the network structure of the dual DQN proposed by Wang et al. (2015) and double DQN structure (Van Hasselt et al., 2016) suggested by Hasselt et al. In addition, to consider non-visual information and temporal information such as speed and yaw rate of the own vessel, we have constructed input layer for the low-dimensional state.

To train the proposed collision avoidance DRL network, we performed repeated RL simulations. During the simulation, experience tuples (s, a, r, s') are collected into the experience buffer until the number of the tuple reaches to maximum buffer size. (in this case, 750,000) In every training steps, a mini-batch of the tuple set is randomly selected and will be used as a training data for policy update. Once the simulation-based training process was complete, the trained DRL network model will be implemented in a real-world platform for validation experiments. Therefore, precise modeling of the simulation environment is a crucial aspect to guarantee the performance of the simulation based on the trained DRL network in a real-world regime. In this regard, we adopted a dynamic model derived from the system identification experiment (Woo et al., 2018) to reflect the maneuvering characteristics of the research target USV platform. In addition, the time delay and the measurement noise of the navigation sensor and sampling rate are considered to improve the reality of the simulation.

In the RL problem, the exploration strategy is one of the most important design factors to increase the effectiveness and stability of the learning process. In this work, we utilized the SMDP model to

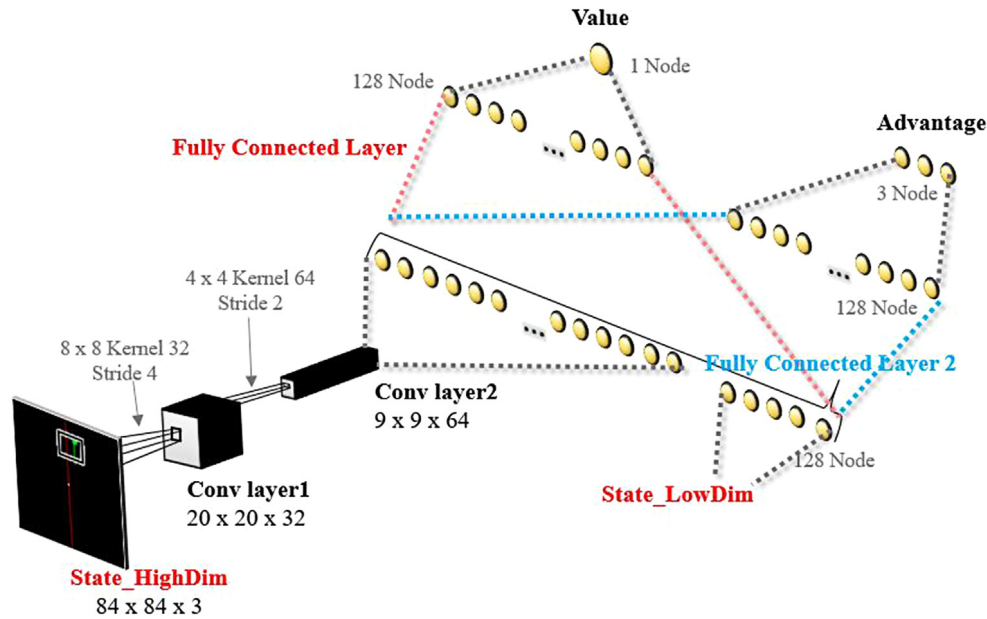


Fig. 6. Structure of the proposed DRL network used for USV collision avoidance decision-making.

handle temporally extended behavior (e.g. path following guidance) instead of instant action (e.g. course angle). As a result of this difference, the conventional exploration strategy that is typically used to solve the MDP problem (such as epsilon-greedy exploration) cannot be directly applied. In the single vessel collision avoidance scenario, a single appropriate avoidance behavior is required to avoid the target obstacle. The purpose of the RL is to determine the appropriate time required for avoidance during the episode. Thus, the exploration strategy should focus on the uniform exploration of avoidance behavior with respect to the temporal period in an episode, instead of the uniform exploration of the action space each time step (as in the case of the epsilon-greedy exploration strategy). To reflect such a difference, we propose an exploration method that is applicable to the DRL based ship collision avoidance problem.

1. Similar to the epsilon-greedy exploration, the epsilon value ϵ is selected. The epsilon value gradually decreases during the progression of the training process.
2. Once an obstacle is classified as an alert obstacle according to the CPA information, the TCPA of the alert obstacle is calculated. ($TCPA_{init}$)
3. Exploration is performed with a probability of ϵ . During the exploration, a TCPA value $TCPA_{exp}$, which is a randomly selected value between 0 to $TCPA_{init}$, is calculated.
4. In the case of the exploration, avoidance behavior is implemented when the $TCPA_{init}$ of the obstacle is less than $TCPA_{exp}$. If there exist multiple avoidance behavior candidates, the direction of the avoidance is randomly selected.

For uniform training of the DRL network in various situations, we composed a single episode of the RL as a combination of six different sub-episodes for different encountered situations. Each sub-episode is in charge of (1) head-on, (2) crossing (give way), (3) crossing (stand on), (4) overtaking, (5) multiple static obstacles and (6) multiple dynamic obstacle encounter situations. The initial position and velocity of the obstacle are randomly selected within a predefined range of corresponding encounter situations.

Fig. 7 describes the training result for the DRL network with respect to the collision avoidance scenario. The x-axis of the figure represents the training time whereas the y-axis represents the episodic reward, which indicates the efficacy of the agent USV in avoiding collisions.

Episodic reward is defined as the sum of each of the six sub-episode rewards. In Fig. 7, the red solid line represents the moving average of the episodic reward (for 200 episodes window), while the green shaded region represents the standard deviation of the episodic reward within the moving window. The training result of Fig. 7 is achieved from a single training process, conducted for approximately 110 h. The number of total training steps is 800,000 steps, and it took about 0.5 s for a single step of the DRL. Based on the training, we can verify that the effectiveness of the DRL network in collision avoidance increases with training. After a certain time period, the slope of the performance index converges to approximately zero and the policy converges to the final policy. The training process of the DRL network is performed using the TensorFlow framework and the Adam Optimizer is used for optimization during training. To train the DRL network, a desktop PC with a 6th generation i7 3.40 GHz CPU, 16 GB RAM, and a GeForce GTX970 GPU was used.

4. Validation

In this chapter, we provide validation results for the proposed collision avoidance method. Before directly implementing the trained policy network for collision avoidance simulation, we attempted to evaluate the validity of the proposed method using several approaches. These included an assessment of the collision risk and the saliency map. After this process, several collision avoidance simulations and full-scale USV experiments were performed for different encountered conditions to evaluate the effectiveness of the proposed method and verify its applicability in the real-world domain.

4.1. Research target USV

In this study, we selected a 14 ft wave adaptive modular vessel (WAM-V) type USV as the research target. A WAM-V USV is catamaran-shaped USV and composed of two pontoon-based hull and upper deck. There are two suspensions between the upper deck and each pontoon, which are designed to minimize wave-induced motion of the upper deck. This USV has a length of 4.88 m and a breadth of 2.5 m. Thrust is produced by two electric motors mounted on each hull and the maximum speed of the vehicle is up to 5 knots. Since the direction of the thrust force is fixed, the vehicle can generate steering control moment by generating RPM differences for the port and

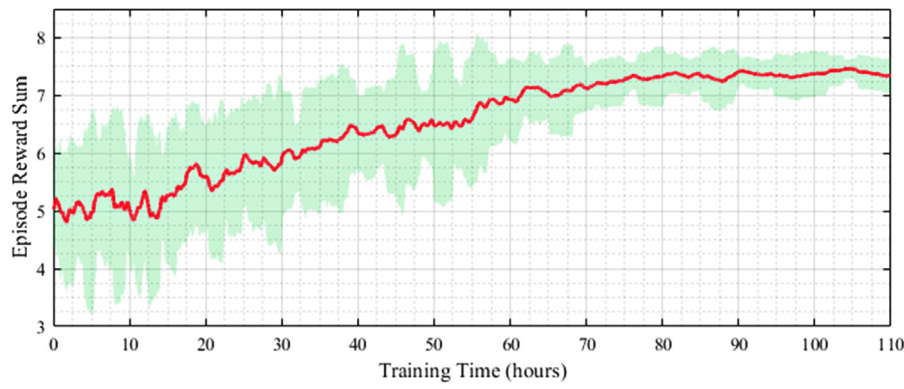


Fig. 7. Episodic reward during the training process of the DRL.

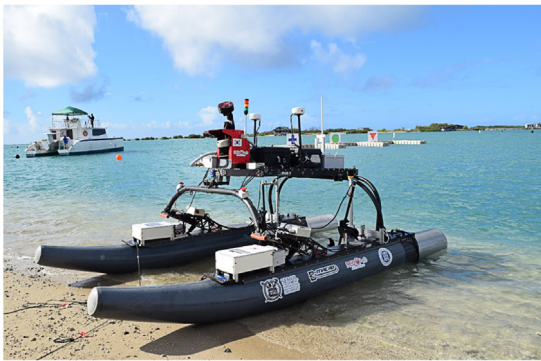


Fig. 8. WAM-V USV platform used in this work. Photo taken from Maritime RobotX Challenge 2016 (ASV competition), Hawaii, U.S.

starboard side thrusters. Fig. 8 shows an image of the WAM-V platform used in this work as the research target vehicle. Woo et al. (2018) recently conducted a dynamic system identification of the identical WAM-V platform. In this work, a dynamic model of the WAM-V USV was derived based on system identification experiments. Thus, for the collision avoidance simulation environment, we used the dynamic system suggested by Woo et al. (2018).

4.2. Collision risk assessment

Once the training process was completed, we utilized the trained network to estimate the collision risk (Hasegawa and Kouzuki, 1987; Ahn et al., 2012) for verification of the situation awareness capability under a given input situation. By definition, the collision risk is an index that describes the negative extent of the encountered situation. This is the opposite definition of the value function in reinforcement learning. A value function is defined as the accumulated future reward. Intuitively, this term describes the positive extent of the encountered situation. Thus, by reversing the sign of the value function and normalizing the term with a maximum value V_{max} as Eq. (13), we define a collision risk CR using the value function $V(s)$ of the reinforcement learning network.

$$CR = \frac{V_{max} - V(s)}{|V_{max} - V_{min}|} \quad (13)$$

For collision risk estimation, we performed virtual experiments by applying arbitrary input data to the DRL network. By analyzing the output value function of the corresponding input situation encountered, the recognition ability of the proposed method can be verified. For the virtual experiment, we defined 5000 encounter situations as input data. In each encountered situation, a single obstacle ship is located. The speed and course angle of the obstacle vehicle are fixed as the negative

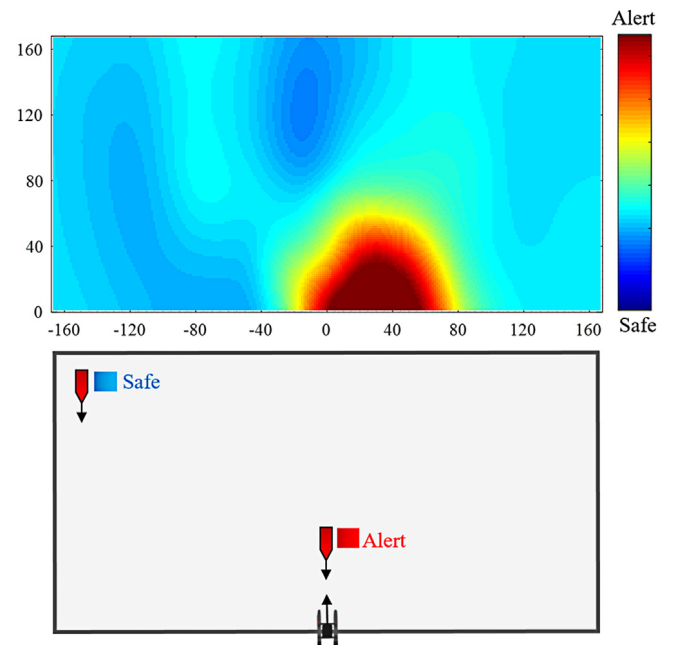


Fig. 9. Collision risk estimation result with respect to the position of the obstacle vessel.

y-axis direction and 1.5 m/s respectively. For the relative position of the obstacle vehicle, the range of the initial lateral position is defined as -168 to 168 , and the variables are divided into 100 equal steps. The range of the initial longitudinal position is set as 0 to 168 and is divided into 50 equal steps. Fig. 9 shows a color representation of the collision risk estimation result with the corresponding obstacle ship location. Based on the virtual experiments, we can determine that the collision risk tends to be high as the distance to the obstacle ship become is reduced. In addition, the proposed method recognizes that an obstacle ship on the port side has a higher collision risk than an obstacle ship on the port side. A similar tendency was also observed in previous works, such as the asymmetry ship domain proposed by Coldwell (1983), Goodwin (1975) and Davis et al. (1980). This tendency is due to the influence of COLREGs. According to COLREGs, in most of the encountered situation, a ship must perform a starboard side turning maneuver to avoid the obstacle vehicle. However, if there is an obstacle on the ship's starboard side, the vehicle tends to be exposed to a more risky situation during collision avoidance. As a result of the iterative learning process, the agent can discover such tendency on its own without prior knowledge.

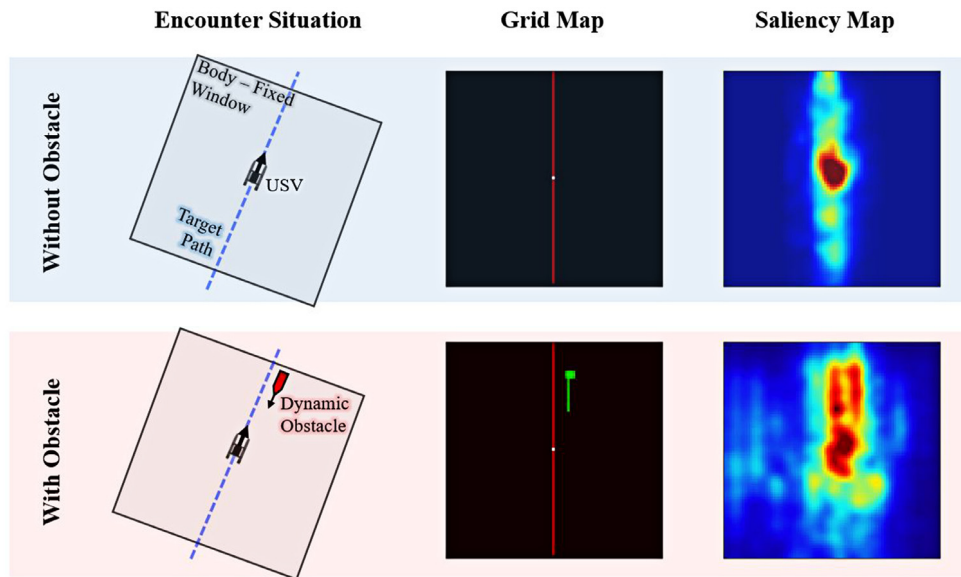


Fig. 10. Saliency map of the encountered situations based on the presence of the obstacle.

4.2.1. Saliency map analysis

As described in Section 3.2.1, we processed maritime traffic encounter situation information to produce visual information using a grid map. This visual information is then used as input data of the convolutional neural network (CNN) layer. When the CNN analyzes the current encounter situation, we can determine the regions of the grid map that the CNN focuses on using a saliency map. Using this tool, we can visualize the active region of the input grid map. Simonyan et al. (2013) proposed a systematic approach for obtaining a saliency map based on the Jacobians of the trained value function of the DRL with respect to the image pixel. If there is a higher Jacobian in a certain region of an image, this implies that visual information in this region has a significant effect on the value function. For the validation purpose, we produced two encounter situation grid maps virtually and used them as input images for the reinforcement learning network. Fig. 10 shows a comparison result for an encounter situation grid map and the corresponding saliency maps. There are two grid maps, one without obstacle and another one with an obstacle near the vessel. In the first case, the reinforcement learning network focuses on the region where the own vehicle is located and the overall (both the forward and behind side) region of the target path. Moreover, once there is an approaching dynamic obstacle in front of the own vehicle, the network tends to focus more on the region where the obstacle is located instead of the behind side region of the target path.

4.3. Collision avoidance simulation

During the collision risk assessment and saliency map analysis, we verified the applicability of the proposed method for the collision avoidance problem. However, to evaluate its effectiveness during the collision avoidance stage, corresponding simulations should be performed. In the collision avoidance simulation, we assumed that the obstacle information is provided by other perception devices.

From the perspective of ship collision avoidance problem, the level of uncertainty tends to be high when there is a large number of obstacles near the vehicle. This is because the ship collision avoidance problem is a decision-making problem in which previous decisions affect future events. As a result, an acceptable avoidance maneuver decision made in the past may have a catastrophic result in the future.

Considering a collision avoidance scenario with multiple ships as described in Fig. 11. Firstly, the USV should avoid an obstacle vehicle location in front in the case of an overtaking encounter situation

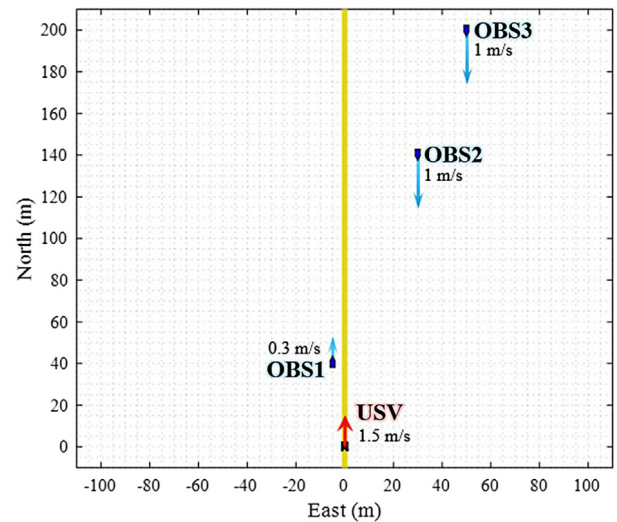


Fig. 11. Example of complex multiple ship encounter scenario.

(OBS1). Behind the OBS1, two obstacle vehicles are approaching toward the USV with head-on encounter situation (OBS2 and OBS3). According to the COLREGs, either port or starboard side avoidance maneuvers are acceptable to avoid the OBS1 (overtaking encounter situation). Since the obstacle ship is located on the port side of the USV, an avoidance maneuver with starboard side turning is desirable considering the relative position of the OBS1 and target path. However, once the USV performs the starboard turning maneuver, the vehicle will encounter OBS2 and OBS3 and it will have to perform successive avoidance maneuvers to avoid a collision. This is a typical example of a decision-making problem. To address these issues, decisions should be made based on the expectation of future events according to the current action choice. Unlike the conventional approach, the proposed DRL based collision avoidance method can foresee future situations using the action value function that was trained via repeated collision avoidance simulations.

For the validation of the proposed method, we performed collision avoidance simulations for the encounter situation described in Fig. 11 using both the proposed method and conventional approach. For the

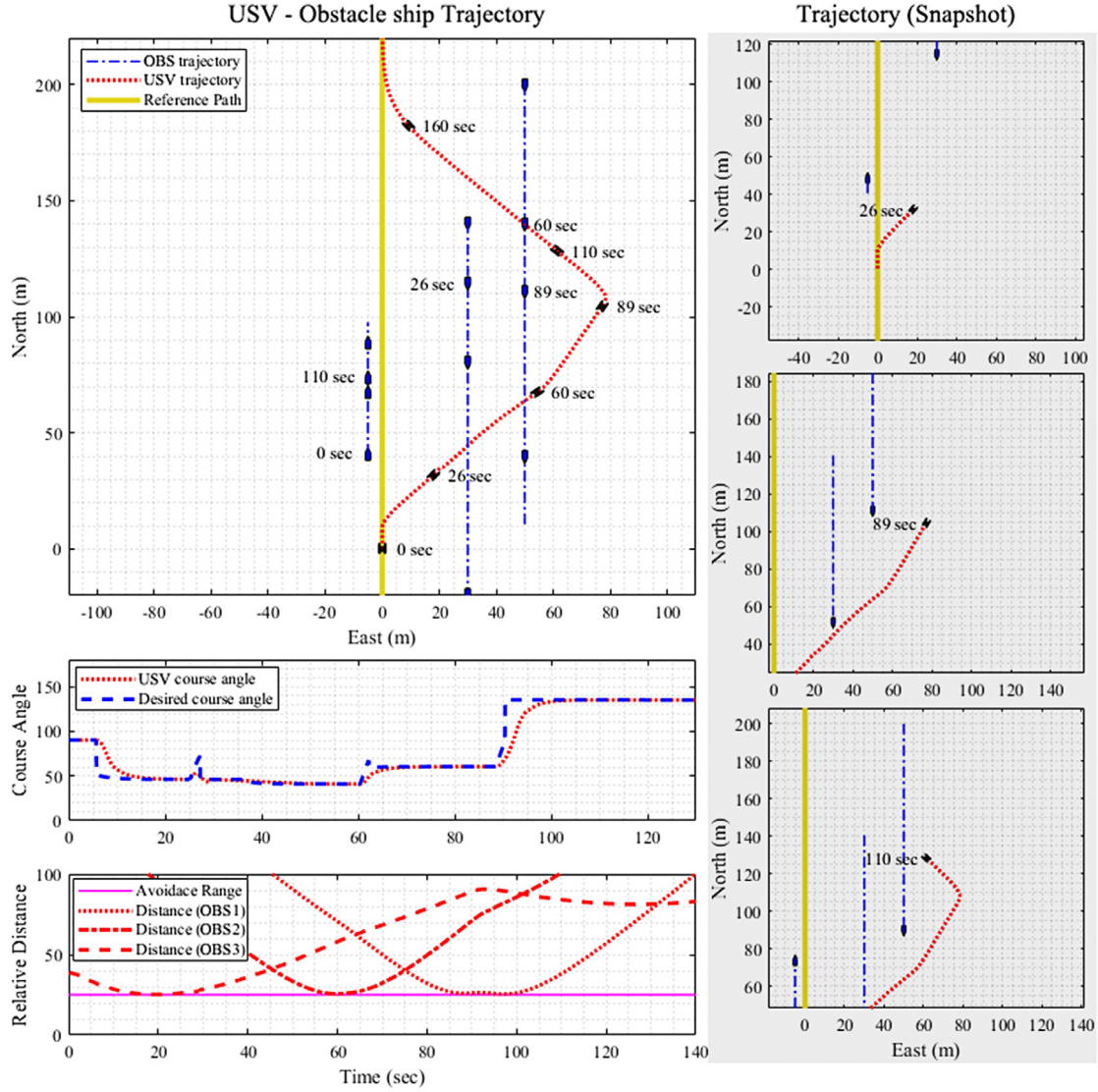


Fig. 12. Simulation result for multiple ship collision avoidance using conventional CPA-based approach.

conventional method, a CPA-based collision risk assessment method suggested by Hasegawa and Kouzuki (1987), and the velocity obstacle method (Kuwata et al., 2014) for path planning were used.

Fig. 12 describes collision avoidance simulation result based on conventional approaches. As the USV faces the overtaking vessel OBS1, the kinematic information of the USV and obstacle vessels are transformed into CPA-based parameters. Using the calculated TCPA and DCPA, an encounter situation was considered as an alert situation at approximately 5 s. When considering the current encounter situation, a starboard side avoidance maneuver is determined using the VO method. Although the VO method can address the path planning of multiple vessels, the effect of OBS2 and OBS3 when considering the avoidance maneuver for OBS1 is limited due to the large distance to OBS2 and OBS3. After avoiding OBS1, the USV encountered OBS2 and OBS3 and successive avoidance maneuvers were performed to avoid collisions.

Fig. 13 represents the collision avoidance simulation results obtained using the proposed method in an identical scenario. In this case, once the USV encounters the OBS1, it can determine the likely outcome if a starboard maneuver is performed to avoid a vehicle based on the current encounter situation grid map. Thus, instead of performing a starboard turning maneuver, the vehicle turned to the port side in order to avoid OBS1. The port side avoidance maneuver seems to be inefficient when considering OBS1 only. However, considering the

encounter situation, the maneuver is a far more efficient and safe avoidance maneuver. Based on experiences during the training stage, the proposed DRL network can develop the ability to understand and interpret complex ship encounter situations on its own.

This ability to predict imminent risk under the complicated encountering situation is one of the most distinctive characteristics of this proposed method. Such ability is one of the most crucial aspects of autonomous ship collision avoidance. This is because most ship collision incidents occur when there is a high level of uncertainty due to complex and ambiguous encounter situations. To deal with such complexity, a high-level decision should be made using the entire complex encounter situation information. However, conventional collision risk assessment methods (both CPA and ship domain-based collision risk assessment methods) cannot provide such information because these approaches tend to simplify the encounter situation. When conventional collision risk assessment interprets complex and ambiguous encounter situation, the encounter situation is parameterized into a few parameters. During the parameterization, most of the delicate and complex information is diminished and diluted into a few parameters. On the contrary, the proposed DRL based method uses the complete information of the current encounter situation to make collision avoidance decisions. Therefore, it can recognize complex encounter situation. Fig. 14 schematically

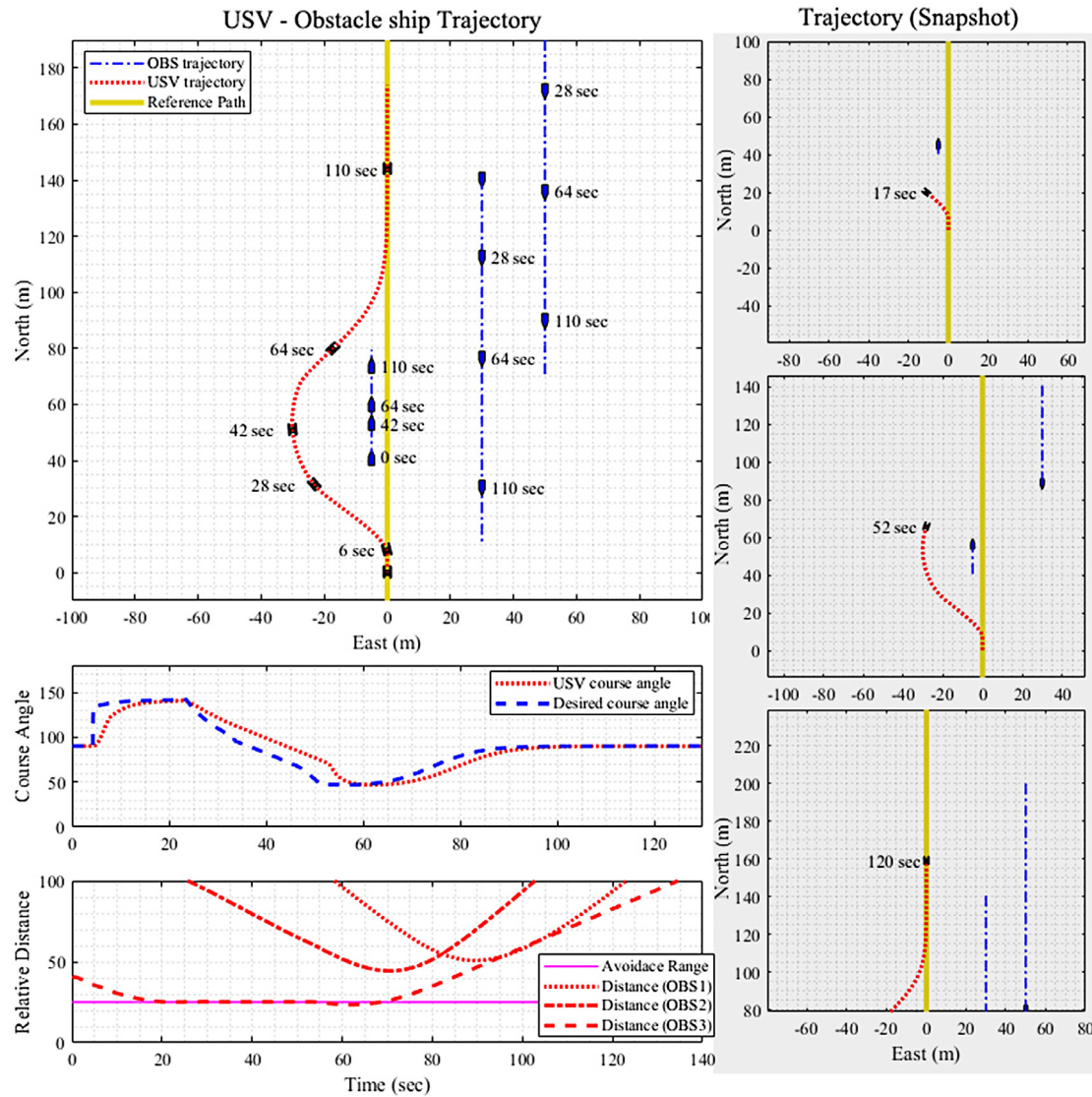


Fig. 13. Simulation result for multiple ship collision avoidance using the proposed DRL method.

illustrates such difference between conventional approaches and the proposed method.

4.4. Collision avoidance experiment

To validate the proposed collision avoidance decision-making method in the real world, we performed free running experiments with a full-scale USV. For the experiment, we utilized the trained DRL model network that was used during the validation simulation and deployed the network into the USV control system. Using the parameters from the extracted DRL network, the action-value function (advantage) of each behavior candidate can be calculated in real-time. For navigation of the USV, we used navigational sensor information such as dual frequency GPS (Novatel Smart6) and an INS sensor (MicroStrain 3DM-GX3). For the obstacle perception, we assumed that there are virtual obstacles in the experimental site to simplify the range of the research. In real-world application, we defined constant velocity obstacle models and calculated the obstacle's kinematic information based on the running time of the experiment. The free running experiment was performed in lake Pyongtaek and the WAM-V USV platform is described in section. 4.1 was used as the target USV platform. For experimental validation of the collision avoidance capability of the proposed method, both the

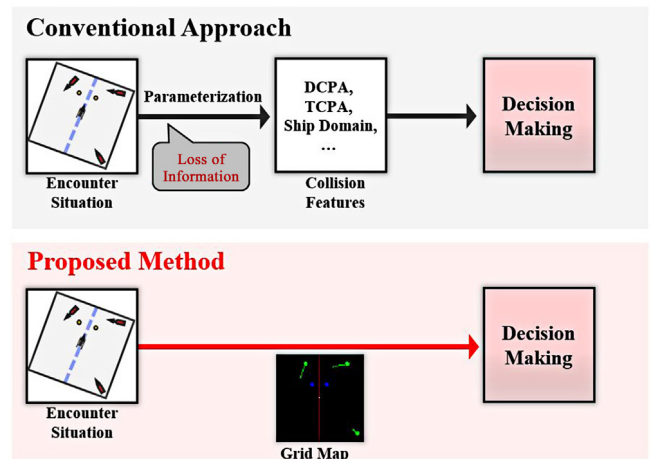


Fig. 14. Comparison between the conventional collision risk assessment approach and the proposed method.

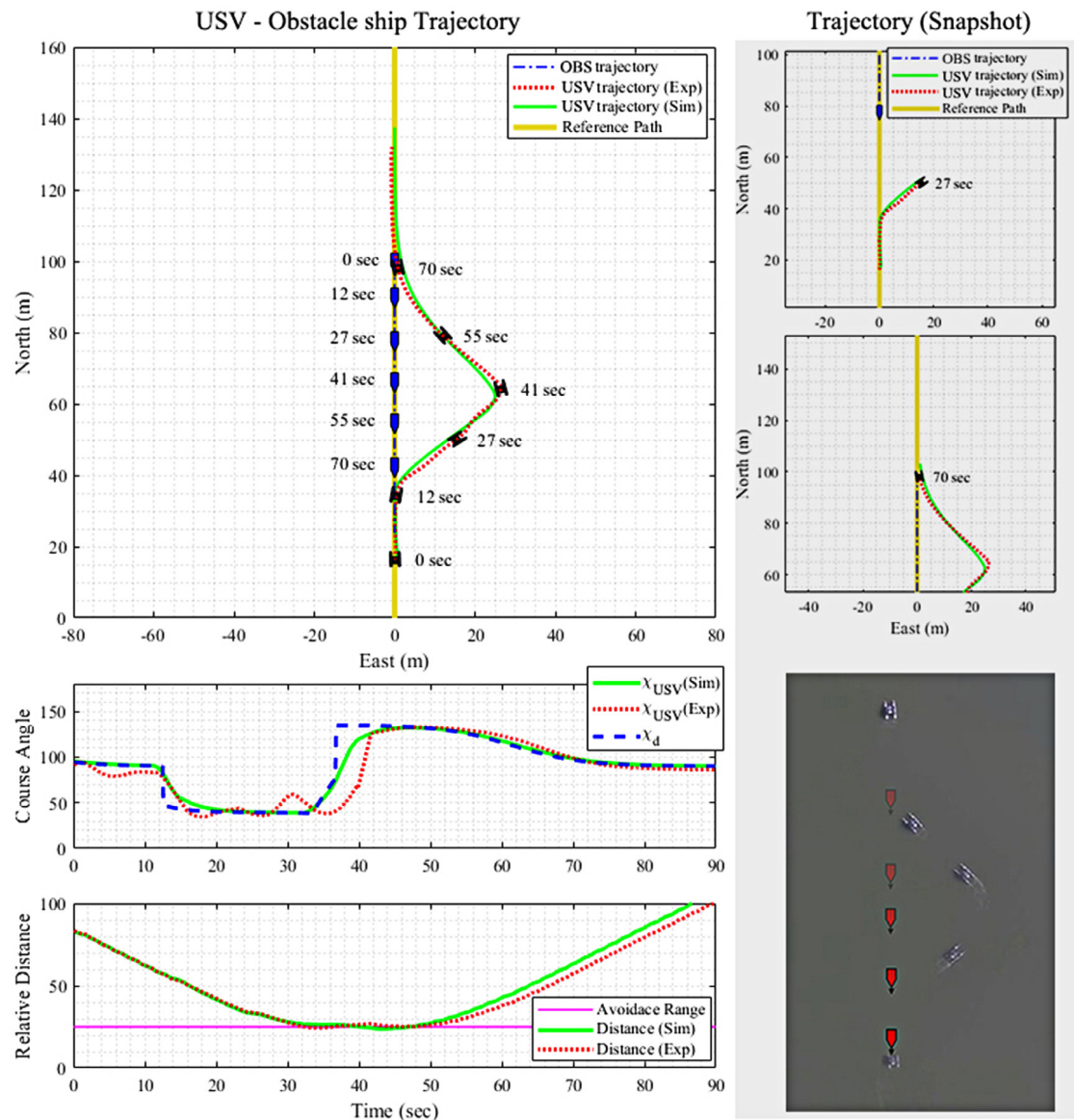


Fig. 15. Collision avoidance experiment and simulation results during the head-on scenario.

single obstacle ship (head-on and overtaking encounter situation) and multiple obstacle ship scenario were selected as experimental scenarios.

4.4.1. Head-on encounter scenario

In the head-on encounter scenario, we defined a head-on encounter situation with an obstacle ship approaching the USV with a ship speed of 0.8 m/s. Fig. 15 represents the experimental result for collision avoidance during the head-on encounter scenario. The red dotted line represents autonomous experimental results and the green solid line represents the corresponding simulation results for identical initial conditions of the USV and obstacle. The avoidance maneuver was performed when the distance between the two vehicles was approximately equal to 65 m and the maximum course angle change was measured as 46°. In the reward function designing process, the desired course angle change was defined as 30°. However, the measured course angle change tends to have a higher value. We concluded that this is due to the different dynamic characteristics of the controller in the USV platform and simulation. Given that the dynamic model identification process performed by Woo et al. (2018) was limited to open loop system identification, the controllers in both the real-world plant and the simulation were independently designed and tuned. In the experimental

results, a higher overshoot level of the tracking error is determined for the course angle at 15 s. At 31 s, oscillatory motion of the course angle occurred. Based on detailed analysis, it was determined that this is because of the modeling uncertainty or environmental disturbances. As a result of the collision avoidance during the experiments and simulations, the USV returned to its target path. In the case of the experimental results, there is a delay of approximately 4 s between the moment of recovery and the target path compared to the result of the simulation. This tendency is due to the repeated oscillatory motion of the course angle. Given that the USV is a differential thruster type vehicle, to produce a steering control moment, the RPM of the port and the starboard side thruster should be changed. As a result of the RPM change, a speed reduction may occur and the recovery time may be delayed. Despite the temporal difference, the USV was able to stably avoid the obstacle ship in both the experiments and simulations with the guarantee of a predefined avoidance range margin of 25 m. Fig. 15(right) shows the snapshot trajectory of the USV and obstacle ship during the head-on encounter collision avoidance experiment, as well as aerial images of the conducted experiment.

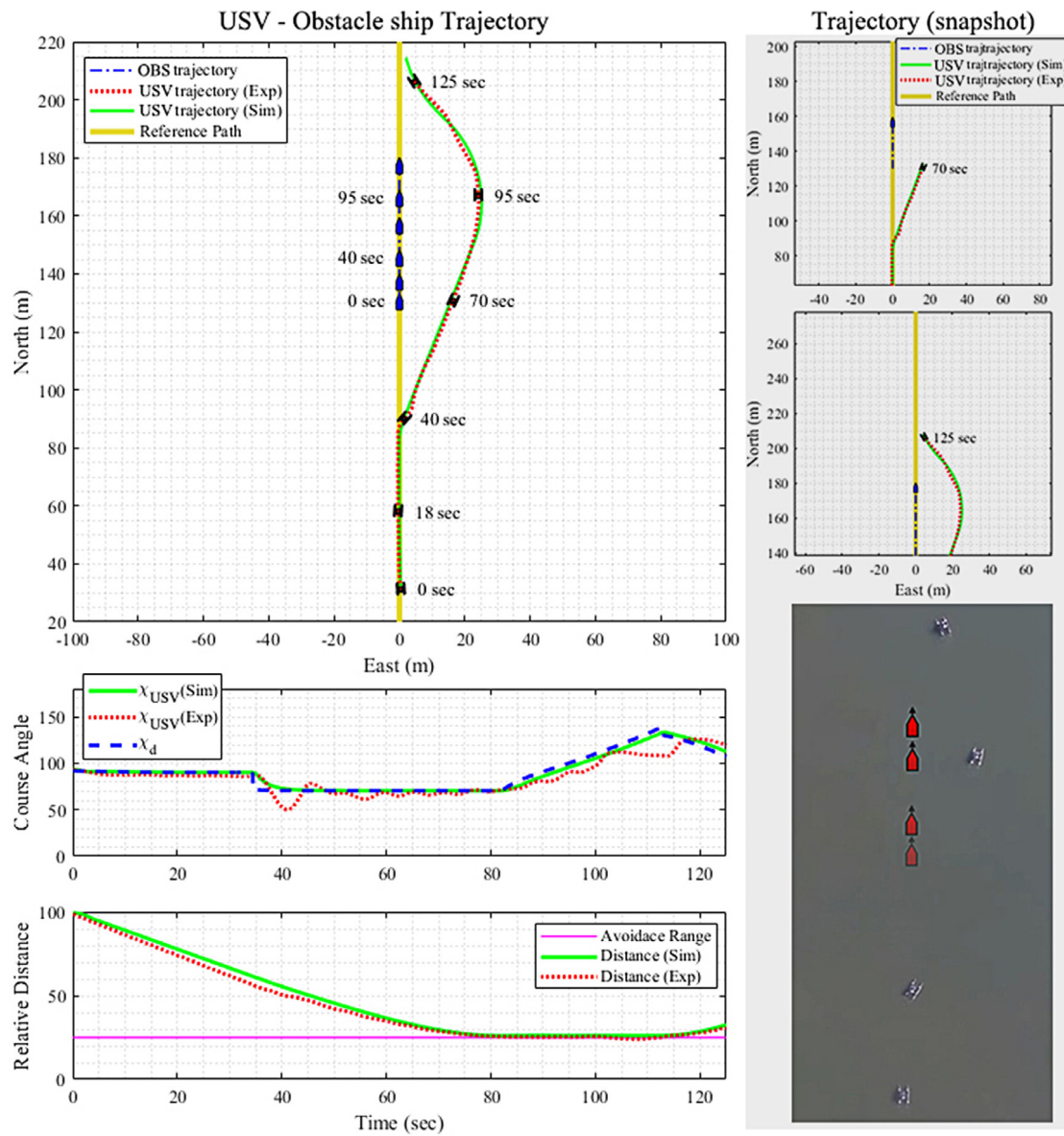


Fig. 16. Collision avoidance experiment and simulation results during the overtaking scenario.

4.4.2. Overtaking encounter scenario

Fig. 16 illustrates the experimental results for the overtaking encounter collision avoidance scenario. For the initial condition, we the obstacle ship was set to head North (90°), which is same as the USV and speed of the obstacle ship and the USV were set as 0.4 m/s and 1.5 m/s, respectively. In the case of the overtaking encounter scenario, the avoidance maneuver was performed over a wide area due to the low relative velocity. Based on the experiments, the course angle overshoot error which was found in the result of head-on scenario can also be determined. However, the USV tends to guarantee a predefined avoidance range margin of 25 m throughout the entire maneuvering process. Due to the slow relative velocity, the simulation and experiment results tend to have a small trajectory error. By examining the trajectory point at 125 s, it was determined that there is 6 m of difference in the USV position between the experiment and simulation. As indicated in the case of the head-on scenario, we determined that this is due to the speed reduction in the experiment caused by the oscillation of the course angle. Fig. 16(right) describes the snapshot trajectory of the USV and

obstacle ship, and the aerial snapshot acquired during the overtaking experiment.

4.4.3. Multiple ship encounter scenario

Fig. 17 illustrates the initial condition for the multiple ship avoidance scenario. In this encounter situation, three obstacle vessels are defined as obstacles of the USV. According to the course angle change described in Fig. 18, the initial avoidance maneuver starts approximately at 10 s in both the experiment and simulation. The USV then returns to the target path at approximately 70 s. At this instance, it is evident that there is 6 m of difference between the position of the USV in the simulation and experiment (Fig. 19). The USV tends to follow the target path for 82 s then encounters the second obstacle ship. At that instance, the USV in the simulation is ahead of the USV in the experiment. Therefore, the former starts to perform avoidance maneuvers relatively early. These differences in the starting time of collision avoidance for different collision avoidance trajectories are as illustrated in the snapshot trajectory (Fig. 19) at 110 s. Once the USV returns

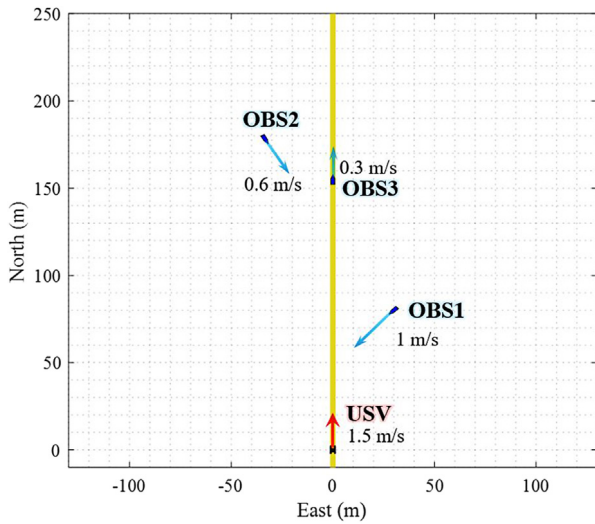


Fig. 17. Multiple ship encounter scenario for collision avoidance experiment and simulation.

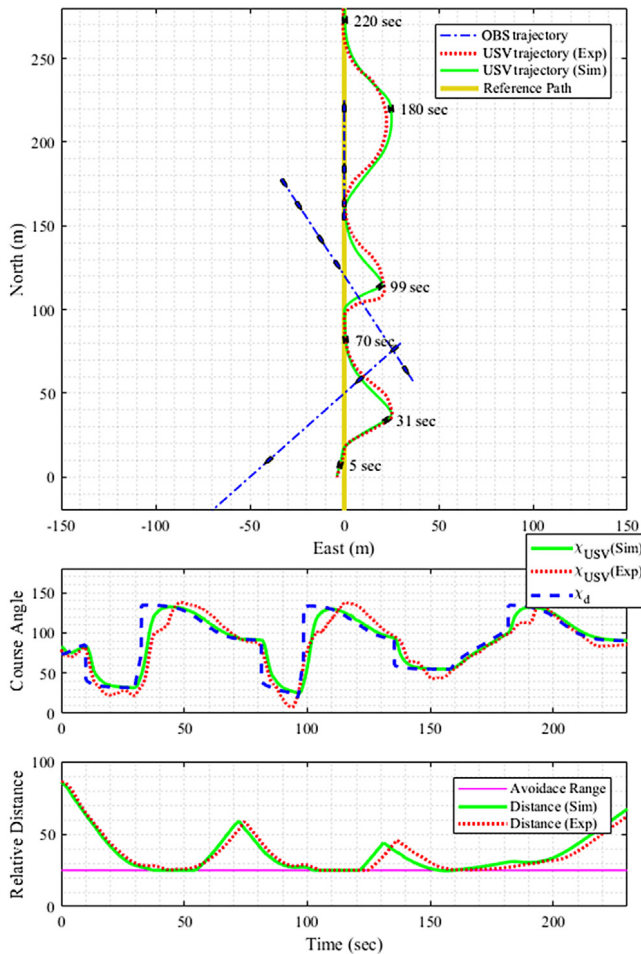


Fig. 18. Collision avoidance experiment and simulation results during the multiple ship avoidance scenario.

to the target path (at approximately 140 s), it encounters an obstacle ship in an overtaking situation. To overtake the vehicle, the USV starts to perform a starboard maneuver. According to the trajectory result described in Fig. 18, it can be inferred that the corresponding avoidance in the simulation was performed at an earlier moment compared to the experiment. Fig. 18 depicts a snapshot of the vehicle's trajectory during each encounter situation of the scenario.

5. Conclusion

In this work, we proposed a DRL-based collision avoidance decision-making algorithm and conducted simulations and experiments to validate its effectiveness. In the proposed method, a USV recognizes its nearby environment using a grid map representation of the encountered situation. The generation of this information using a visual grid map representation is one of the crucial aspects of the proposed method. By providing information on the convolutional neural network (CNN), the feature extraction capability of the CNN can be adapted to develop an intuition for collision situation awareness. To consider COLREGs in the process of the collision avoidance decision, we designed reward functions for the DRL network with respect to each encountered situation. The resulting DRL network was trained based on repeated collision avoidance simulations. It is therefore able to reflect experiences during various encounter circumstances. As a result of this work, we can state the following conclusions.

- According to the collision risk assessment analysis, it was determined that the proposed method can identify the risk of collision in a similar manner to conventional approaches. For example, the proposed method recognized the asymmetric characteristic of the collision risk (highlighted in previous works Coldwell, 1983; Goodwin, 1975; Davis et al., 1980) via repeated collision avoidance simulations.
- The proposed method can recognize complex and ambiguous situations with multiple obstacles. In addition, it can empirically foresee future risk of collision when considering avoidance action candidates.
- The proposed method can select a suitable avoidance strategy according to the type of encounter. Based on the COLREGs, there are a variety of avoidance requirements for different encountered situations. The proposed method can consider such difference by independently designing the reward function.
- Throughout the full-scale USV collision avoidance experiments under various scenarios, the applicability and the effectiveness of the proposed method in the real-world domain was verified

Although there are several merits, there are also a few limitations of the proposed system. Given that the proposed method repeatedly performs reinforcement learning on a high dimensional state (grid map visual information), this requires hundreds of hours of training. In addition, the application of the trained DRL network is affected by the uncertainty of the modeling of the vehicle because the training process is performed in the simulation environment. If the USV can be trained in a real-world environment, this limitation will be overcome.

For future work, we are planning to consider the effect of environmental load in making collision avoidance decisions in order to increase the reliability and effectiveness of the proposed method in a real-world domain. In addition, we are considering extension of the action space to ship's speed control, as well as steering control. This approach will be effective, especially under the multi-vessel encounter scenario.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

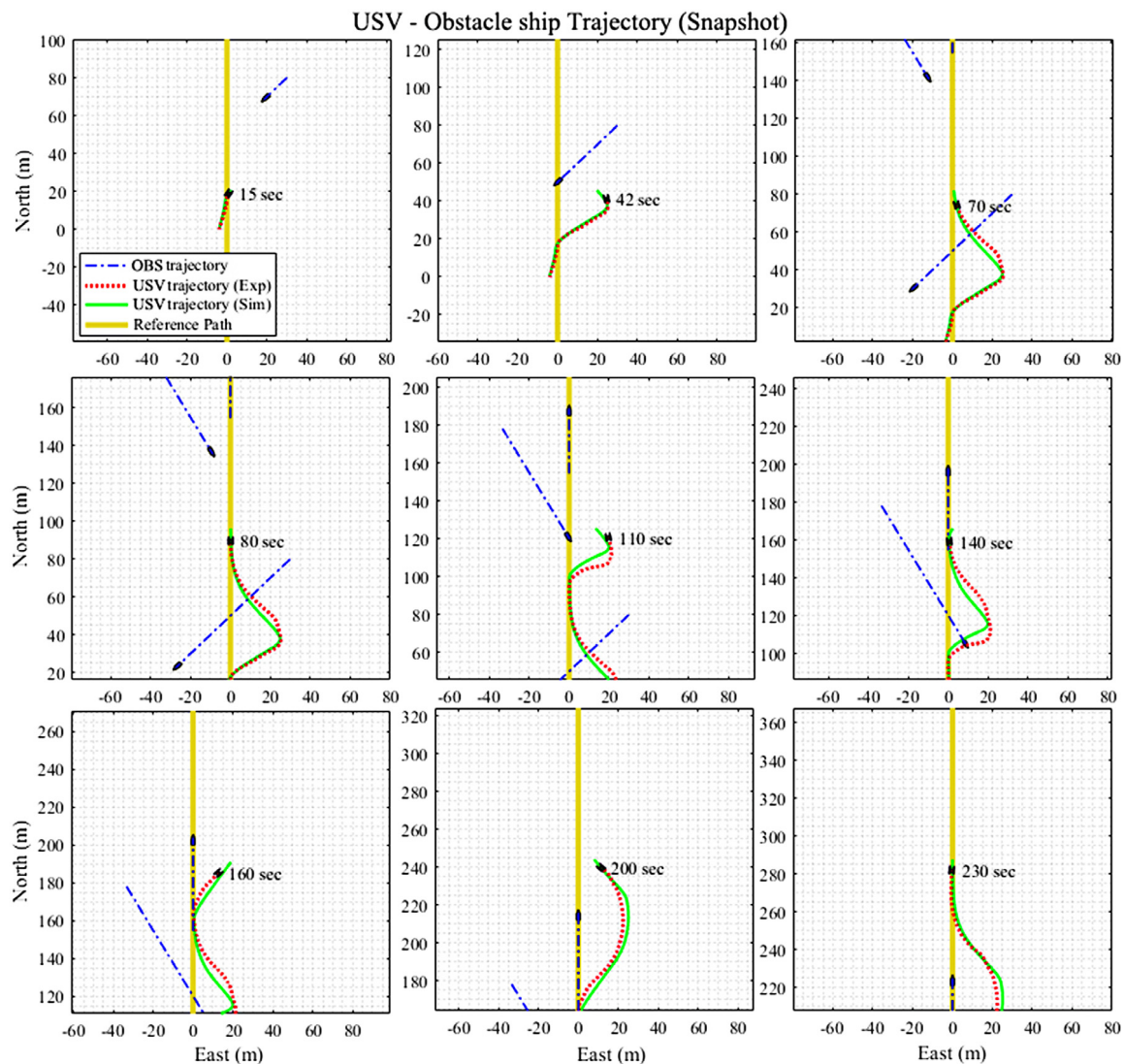


Fig. 19. Snapshot of the collision avoidance experiment and simulation results during the multiple ship avoidance scenario.

CRedit authorship contribution statement

Joohyun Woo: Conceptualization, Methodology, Software, Writing - original draft, Writing - review & editing. **Nakwan Kim:** Supervision, Project administration.

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